

Healthcare Data Analysis Dashboard – Report

Name:Arni G
BSC DS & AI

1. Introduction

The objective of this project is to analyze healthcare data using Power BI and generate meaningful insights related to patient details, hospital admissions, and billing information. The dashboard helps in understanding patterns and supports data-driven decision-making.

2. Data Loading

The dataset was loaded into Power BI using the **Get Data → CSV option**. The data was imported successfully and prepared for transformation.

healthcare_dataset.csv

File Origin: 1252: Western European (Windows) Delimiter: Comma Data Type Detection: Based on first 200 rows

Name	Age	Gender	Blood Type	Medical Condition	Date of Admission	Doctor	Hospital	In
Bobby JacksOn	30	Male	B-	Cancer	31-01-2024	Matthew Smith	Sons and Miller	BL
LesLie TERy	62	Male	A+	Obesity	20-08-2019	Samantha Davies	Kim Inc	M
DaNnY sMith	76	Female	A-	Obesity	22-09-2022	Tiffany Mitchell	Cook PLC	Ac
andrEw waTtS	28	Female	O+	Diabetes	18-11-2020	Kevin Wells	Hernandez Rogers and Vang,	M
adriENNE bEll	43	Female	AB+	Cancer	19-09-2022	Kathleen Hanna	White-White	Ac
EMILY JOHNSOn	36	Male	A+	Asthma	20-12-2023	Taylor Newton	Nunez-Humphrey	Un
edwArD EDWaRDs	21	Female	AB-	Diabetes	03-11-2020	Kelly Olson	Group Middleton	M
ChRisTinA MARTinez	20	Female	A+	Cancer	28-12-2021	Suzanne Thomas	Powell Robinson and Valdez,	Ci
JASmiNe aGullaR	82	Male	AB+	Asthma	01-07-2020	Daniel Ferguson	Sons Rich and	Ci
ChRISToPher BerG	58	Female	AB-	Cancer	23-05-2021	Heather Day	Padilla-Walker	Un
mlchElLe daniELs	72	Male	O+	Cancer	19-04-2020	John Duncan	Schaefer-Porter	M
aaRon MARTiNeZ	38	Female	A-	Hypertension	13-08-2023	Douglas Mayo	Lyons-Blair	M
connOR HANsEn	75	Female	A+	Diabetes	12-12-2019	Kenneth Fletcher	Powers Miller, and Flores	Ci
rObErT bAuer	68	Female	AB+	Asthma	22-05-2020	Theresa Freeman	Rivera-Gutierrez	Un
bROOkE brady	44	Female	AB+	Cancer	08-10-2021	Roberta Stewart	Morris-Arellano	Un
MS. nAtalie gAMble	46	Female	AB-	Obesity	01-01-2023	Maria Dougherty	Cline-Williams	BL
haley perkins	63	Female	A+	Arthritis	23-06-2020	Erica Spencer	Cervantes-Wells	Un
mRS. jamiE cAMPBELl	38	Male	AB-	Obesity	08-03-2020	Justin Kim	Torres, and Harrison Jones	Ci
LuKE BuRgEss	34	Female	A-	Hypertension	04-03-2021	Justin Moore Jr.	Houston PLC	BL
dANIEL schmidt	63	Male	B+	Asthma	15-11-2022	Denise Galloway	Hammond Ltd	Ci

Extract Table Using Examples Load Transform Data Cancel

3. Data Cleaning

Data cleaning was performed in Power Query Editor:

- Missing values were handled
- Data types were corrected
- Duplicate records were removed
- Unnecessary columns were removed
- Text values were standardized

Name	Age	Gender	Blood Type	Medical Condition	Date of Admission	Doctor	Hospital	Insurance Provider
Laura Myers	61	Male	B+	Obesity	12 March 2021	Kyle Williams MD	Brennan Bentley, and Hunter	Cigna
William Rogers	84	Male	A-	Asthma	31 August 2020	Heather Petty	Robinson-Ritter	Cigna
Jason Smith	74	Male	A-	Obesity	03 September 2022	Diana Martinez	Mcknight Perry Rivera, and	Cigna
Rachel Bush	61	Female	O-	Hypertension	27 March 2020	Adrian Vaughn	Lopez Ingram, and Barber	Cigna
Nicole Perry	83	Female	O-	Asthma	08 February 2020	Susan Miranda	Inc Soto	Cigna
Donald Simon	66	Male	B+	Hypertension	27 November 2023	James Hendrix	Morris and Sons	Cigna
Carlos Edwards	73	Male	B-	Hypertension	07 September 2022	Nichole Salas	Vaughn-Williamson	Cigna
Mrs. Jennifer Chandler	77	Female	B-	Asthma	27 November 2020	Jessica Schroeder	Willis PLC	Cigna
Stephen Holt	84	Male	O+	Cancer	23 June 2023	Richard Bailey	Chavez, and Smith Hawkins	Cigna
Jose Rivas	81	Female	AB-	Asthma	17 February 2022	Shannon Meyer	Durham, Burch and Webb	Cigna
Jamie Kennedy	62	Male	O+	Diabetes	07 January 2020	Adam Bradley	Frye Inc	Cigna
Alexander Jordan	84	Male	O-	Hypertension	23 April 2023	Mario Gonzales	Garcia-Jacobs	Cigna
Holly Harrell	71	Male	O-	Diabetes	13 June 2022	Kimberly Campbell	Santiago Sons and	Cigna
Angela Leonard	80	Male	AB-	Obesity	20 January 2021	James Goodwin	Kline, Davis Thomas and	Cigna
Brent Gibson	63	Male	O+	Asthma	01 November 2019	Christopher Higgins	Lee, Martinez Hart and	Cigna
Lori Hughes	73	Male	B-	Hypertension	06 March 2020	Janice Hancock	LLC Barnes	Cigna
Mr. Nicholas Garcia	64	Female	A-	Diabetes	03 July 2020	Mathew Williams	Lopez, and Conner Evans	Cigna
Sharon Castro	81	Female	B-	Diabetes	21 March 2024	Holly Harris	and Williams Day, Tyler	Cigna
Christopher Clark	70	Female	AB-	Arthritis	25 July 2019	Jordan Haney	and Greene, Love Gonzalez	Cigna
Benjamin Brown	62	Female	B-	Hypertension	03 March 2021	Samantha Bradford	Rogers PLC	Cigna
Alexander Miller	70	Female	AB-	Asthma	05 February 2023	Michael Buchanan	and Thomas Lopez Lamb,	Cigna
Michael Gillespie	85	Female	O+	Asthma	23 January 2024	Anthony Robertson	Gonzalez Roman, Orozco and	Cigna
Faith Mathis	83	Female	B-	Arthritis	11 April 2021	Adam Bradley	Anderson Group	Cigna
Juan Sharp	67	Female	O-	Asthma	20 January 2021	Kathy Johnson	Larson-Sanchez	Cigna
Jordan Bridges	67	Female	A+	Asthma	19 January 2023	Michelle Johnson	Gordon-Washington	Cigna
Colleen Adams	84	Female	AB+	Cancer	16 March 2020	Scott Hunter	and Holmes McBride Landry,	Cigna

4. Feature Engineering – Age Group Creation

A new column called **Age_Group** was created using Conditional Column:

- Age < 18 → Child
- Age ≤ 35 → Young Adult
- Age ≤ 60 → Adult
- Else → Senior

Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name

	Column Name	Operator	Value		Output
If	Age	is less than	18	Then	Child
Else If	Age	is less than or equ...	35	Then	Young adult
Else If	Age	is less than or equ...	60	Then	Adult

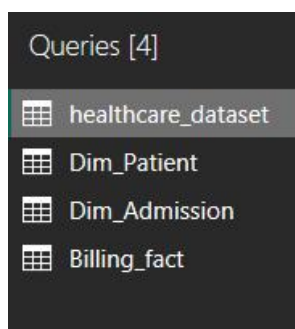
Else
Senior

5. Data Modeling (Star Schema)

The dataset was structured into dimension and fact tables:

Dim_Patient

- PatientID
- Name
- Age
- Gender
- Blood Type
- Age_Group



Dim_Admission

- AdmissionID
 - Admission Type
 - Doctor
 - Hospital
 - Medical Condition
 - Room Number
-

Fact_Billing

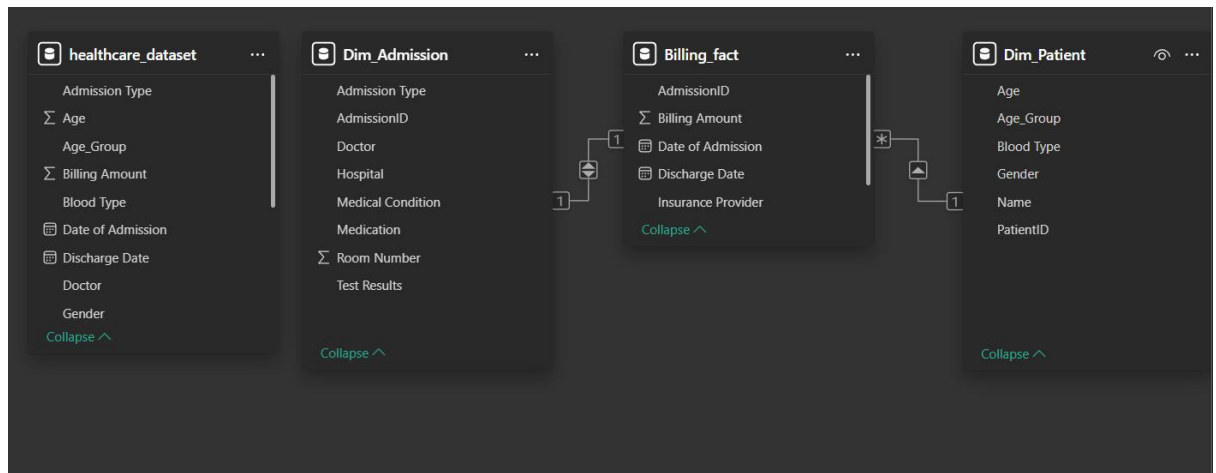
- PatientID
- AdmissionID
- Billing Amount
- Insurance Provider
- Date of Admission
- Discharge Date

Date of Admission ▼	Insurance Provider ▼	Billing Amount ▼	Discharge Date ▼	PatientID ▼	AdmissionID ▼
28 December 2021	Cigna	45820.4627215946	07 January 2022	8	8
01 July 2020	Cigna	50119.2227915485	14 July 2020	9	9
12 December 2019	Cigna	43282.2833577043	28 December 2019	13	13
08 March 2020	Cigna	17440.4654441247	02 April 2020	18	18
15 November 2022	Cigna	23762.2035790596	22 November 2022	20	20
03 January 2021	Cigna	32643.2993532772	16 January 2021	29	29
18 August 2019	Cigna	47909.1288098749	01 September 2019	31	31
12 March 2022	Cigna	48290.6934073642	29 March 2022	39	39
15 September 2021	Cigna	49943.2784987873	05 October 2021	41	41
27 January 2020	Cigna	35633.9554543878	09 February 2020	42	42
24 June 2022	Cigna	25966.328610221	18 July 2022	43	43
01 December 2021	Cigna	29615.4418636462	09 December 2021	47	47
04 May 2020	Cigna	36992.2736882839	13 May 2020	48	48
19 July 2019	Cigna	16320.2324472719	29 July 2019	50	50
17 August 2023	Cigna	19973.4629261596	11 September 2023	55	55
09 April 2023	Cigna	10300.6573113759	21 April 2023	57	57
03 August 2022	Cigna	5517.39364222399	10 August 2022	64	64
07 January 2023	Cigna	49402.2983725271	27 January 2023	68	68
28 December 2023	Cigna	10709.6903889002	02 January 2024	73	73
30 January 2021	Cigna	2305.13913953849	04 February 2021	78	78
30 May 2020	Cigna	22841.3638769057	14 June 2020	84	84
16 May 2023	Cigna	39476.94751438	01 June 2023	89	89
06 September 2022	Cigna	23684.5254727448	27 September 2022	94	94
13 November 2021	Cigna	25425.7278626071	21 November 2021	101	101
13 November 2021	Cigna	25425.7278626071	21 November 2021	101	51137
30 September 2019	Cigna	3879.08460181339	15 October 2019	107	107
14 May 2021	Cigna	37734.742180387	06 June 2021	109	109

Billing_fact (55,485 rows)

Relationships

- Dim_Patient → Fact_Billing (PatientID)
- Dim_Admission → Fact_Billing (AdmissionID)



8. Insights

- Majority of patients belong to **Adult and Senior categories**
- Billing amount is highest for certain insurance providers
- Average hospital stay is around **15 days**
- Some medical conditions have higher readmission rates

9. Conclusion

The Healthcare Data Analysis Dashboard provides a clear and interactive way to analyze patient and billing data. It helps identify trends and patterns effectively. Power BI tools like Power Query, Data Modeling, and Visualization were used to build this dashboard successfully.
